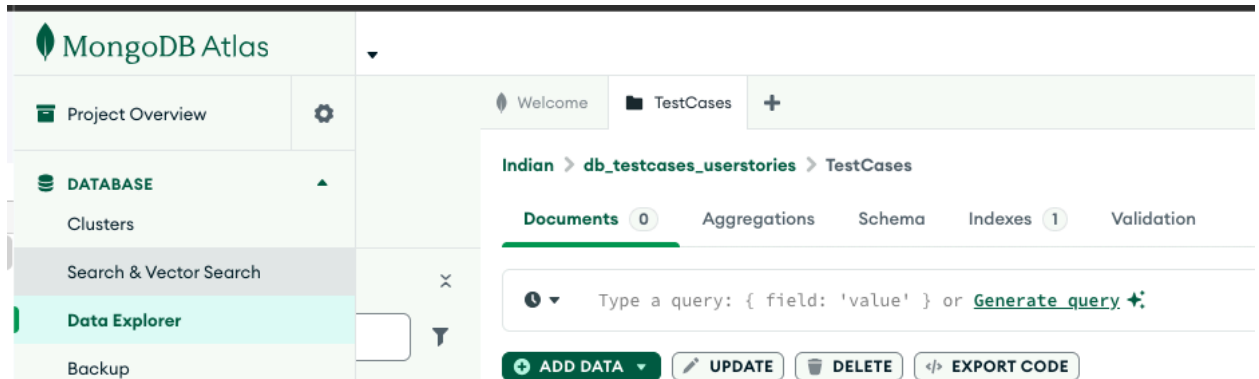


Assignment: Retrieval Pipeline - own userstories in current application

MongoDB Setup

1. Either create new DB or reuse existing DB. We are using existing DB and collections
2. Clear the existing test cases and user stories in the database.
3. Creating Atlas search -> From project overview select 'Search & Vector search'



Type: Atlas search

Index name: bm25_TestCase

Database and Collection: select the DB and collection cleared above.

Configuration Method: JSON Editor

4. Click on Next. In the JSON editor paste the content from "bm25-search.js" file in the rag-mongo-demo-v9 repo. And click Next
5. Click 'Create search Index'
6. Update the .env file in the vs editor
7. Run the terminal run "npm run dev".
8. An UI will open with url "http://localhost:3000/". Configure the settings in the UI to match that of .env file.

Importing test cases and user stories from frontend

1. Select Data Type as Test case or user stories and upload the necessary file. Convert the file for both test cases and user stories.
2. Click on Embeddings and Store in the navigation pane
3. Sort by 'Modified' to get the latest file at top.
4. Select the latest files that were converted above and click on 'Create Embeddings'.
5. Once processed look into the respective collection that the test cases and user stories are available in the MongoDB.

Backend configuration

1. Update the dictionary.js and abbreviationmapping.js file to include the abbreviations that we can expect from the test cases and user stories.
2. Re-run npm run dev

Query Preprocessing & Search

1. In the frontend ui select 'Query preprocessing' in the navigation tool
2. Enter value for query preprocessing - Example : *'reset password using otp'*.
3. Click preprocess query

Query Preprocessing & Search

Transform your query through normalization, abbreviation expansion, and synonym generation before searching

Quick Examples:

UHID patient login issue OTP not working

TC_027 merge UHID records

doctor appointment booking IP admission

password reset OTP verification

ER patient registration BP monitoring

PREPROCESS QUERY

PIPELINE STEPSABBREVIATIONSYNONYM VARIATIONS

Query Transformation Pipeline

ORIGINAL QUERY

"reset password using otp"

NORMALIZATION

"reset password using otp"

4 tokens

ABBREVIATION EXPANSION1 expansions

"reset password using one-time password"

Expanded:

otp → one-time password

SYNONYM EXPANSION5 variations

Multiple query variations generated for comprehensive search coverage

1. "reset password using one-time password"

2. "change password using one-time password"

3. "update password using one-time password"

+ 2 more variations

Processing Metadata

Processing Time	Token Count	Abbreviations	Synonyms
2ms	4	1	3

How Query Preprocessing Works

1. Normalization: Converts text to lowercase, removes special characters, extracts test case IDs

2. Abbreviation Expansion: Expands healthcare abbreviations (UHID → unique health id, OTP → one time password)

3. Synonym Expansion: Generates multiple query variations using synonyms (patient → customer/user/individual)

4. Search Execution: Uses all variations to perform comprehensive search with better recall

Performing vector search

1. In the frontend ui select 'Vector search in the navigation tool
2. Under search query text box enter query 'reset password using otp' and click search

Home > Vector Search

Seman

Q

Query Search

Search through your test cases using semantic similarity with AI embeddings.

Q

Search Query

Enter your search terms and find similar test cases

Enter your search query

reset password using otp

Advanced Metadata Filters

SHOW FILTERS

Results Limit

5 Results

Use descriptive terms related to test functionality

Q

SEARCH

✓

Search Query:

reset password using otp

5 results

Performance:

\$0.000001 cost

7 tokens

Search Results

5 results ranked by semantic similarity

Test Case	Similarity	Description	Source
Reset Password OTP Token Generation	91.2%	Ensure password reset triggers an immediate temporary passcode token via email.	
Auth			
Expired OTP Password Validation	89.1%	Verify that an expired password token fails validation during security resets.	
Auth			
Invalid Password Login Drop	85.0%	Validate that authentication drops if the password does not match system logs.	
Auth			
MFA Verification Code Validation	85.0%	Ensure multi-factor authentication (MFA) layer challenges logins successfully.	
Auth			
Account Lockout Threshold	84.7%	Verify account locks out temporarily after 5 sequential invalid sign-in attempts.	
Auth			

Rows per page: 10 1-5 of 5

Detailed View

Reset Password OTP Token Generation

Score: 0.9116

Auth

Expired OTP Password Validation

Score: 0.8911

Auth

Invalid Password Login Drop

Score: 0.8503

Auth

MFA Verification Code Validation

Score: 0.8497

Auth

Account Lockout Threshold

Score: 0.8470

Auth

Performing BM search

1. In the frontend ui select 'BM25 search in the navigation tool
2. Under search query text box enter query 'reset password using otp' and click search

[Home](#) > [BM25 Search](#)Ks

BM25 Keyword Search

Fast keyword-based search using BM25 algorithm. Best for exact matches, IDs, and specific terms.

Result... 10SEARCH

[SHOW FILTERS](#)

 **BM25 Search Results:** Found 8 test cases in 74ms
Search Type: bm25

Search Results (8)

#1

ECO_TC_007

Score: 10.49

Reset Password OTP Token Generation

Module: Auth

High

Risk: High

Automated

Description: Ensure password reset triggers an immediate temporary passcode token via email.

Steps: 1. Click Forgot Password link 2. Provide active user profile email 3. Click Send Token

Expected Results: A one-time password (OTP) notification code is generated and transmitted to user mail.

 Created: 30/05/2026

#2

ECO_TC_008

Score: 7.20

Expired OTP Password Validation

Module: Auth

Medium

Risk: Medium

Automated

Description: Verify that an expired password token fails validation during security resets.

Steps: 1. Trigger password recovery token 2. Wait for standard 10-minute expiry window 3. Enter the expired code on reset screen

Expected Results: System rejects the token string and prompts the user to request a fresh link.

 Created: 30/05/2026

#3

ECO_TC_005

Score: 3.00

Invalid Password Login Drop

Module: Auth

High

Risk: Medium

Automated

Description: Validate that authentication drops if the password does not match system logs.

Steps: 1. Open sign-in panel 2. Enter valid email address 3. Enter incorrect password string 4. Click Submit

Expected Results: Access is denied with a visible credential mismatch error notification.

 Created: 30/05/2026

#4 ECO_TC_004 Score: 2.39

Successful Buyer Login

Module: Auth High Risk: High Automated

Description: Confirm a registered buyer can authenticate cleanly using core credentials.

Steps: 1. Open login window 2. Enter valid username and password 3. Click Login

Expected Results: User is successfully authenticated and auth tokens are assigned to active session.

Created: 30/05/2026

#5 ECO_TC_003 Score: 2.31

Password Strength Boundary Validation

Module: Auth Medium Risk: Medium Automated

Description: Ensure the registration module enforces password complexity metrics.

Steps: 1. Go to register tab 2. Input an weak 4-character passcode 3. Attempt to submit form

Expected Results: Validation error triggers demanding at least 8 characters including symbols.

Created: 30/05/2026

#6 ECO_TC_002 Score: 2.19

Registration Duplicate Email Prevention

Module: Auth High Risk: High Automated

Description: Verify system blocks registration attempts using an already registered email profile.

Steps: 1. Navigate to sign up page 2. Enter an existing registered email 3. Fill in password fields 4. Click Register

Expected Results: System displays an error message stating email account is already in use.

Created: 30/05/2026

#7 ECO_TC_001 Score: 1.68

Successful User Sign Up

Module: Auth High Risk: High Automated

Description: Validate that a new user can successfully create an account with valid email and password.

Steps: 1. Navigate to the registration page 2. Enter a unique email address 3. Input a valid strong password 4. Click Register

Expected Results: The account should be created successfully and user redirected to the home screen profile.

Created: 30/05/2026

#8 ECO_TC_050 Score: 0.83

User Account Deletion Verification

Module: Auth High Risk: High Automated

Description: Confirm deleting an account purges user profiles completely out of active directories.

Steps: 1. Access secure user profile settings privacy option 2. Submit account deletion confirmation password instructions

Expected Results: Profile state changes to deleted user record is purged out of active directories and session closes.

Created: 30/05/2026

Performing Hybrid search - BM25 + Vector

1. In the frontend ui select 'Hybrid Search' in the navigation tool
2. Under search query text box enter query 'reset password using otp' and click search

<

RAG Pipeline

Home > Hybrid Search

Hybrid Search (BM25 + Vector)

Combines BM25 keyword matching with semantic vector search for best results. Adjust weights to control the balance.

Search Query

reset password using otp

Result

10

SEARCH

Search Weights

BM25 Weight: 50%

Vector Weight: 50%

0% 50% 100%

0% 50% 100%

BALANCED (50/50)

KEYWORD HEAVY (70/30)

SEMANTIC HEAVY (30/70)

SHOW FILTERS

Hybrid Search Results: Found 10 test cases in 942ms

Stats: 8 in both indexes, 0 BM25 only, 2 vector only

Timing: BM25: 54ms, Vector: 886ms

Cost: \$0.000001 (7 tokens)

Search Results (10)

#1 ECO_TC_007 Hybrid: 95.6% Found in: both

Reset Password OTP Token Generation

Module: Auth High Risk: High Automated

Description: Ensure password reset triggers an immediate temporary passcode token via email.

Steps: 1. Click Forgot Password link 2. Provide active user profile email 3. Click Send Token

Expected Results: A one-time password (OTP) notification code is generated and transmitted to user mail.

Score Breakdown:

BM25: 100.0% (raw: 10.49) Vector: 91.2% (raw: 0.91) Hybrid: 95.6%

Created: 30/05/2026

#2 ECO_TC_008 Hybrid: 78.9% Found in: both

Expired OTP Password Validation

Module: Auth Medium Risk: Medium Automated

Description: Verify that an expired password token fails validation during security resets.

Steps: 1. Trigger password recovery token 2. Wait for standard 10-minute expiry window 3. Enter the expired code on reset screen

Expected Results: System rejects the token string and prompts the user to request a fresh link.

Score Breakdown:

BM25: 68.6% (raw: 7.20) Vector: 89.1% (raw: 0.89) Hybrid: 78.9%

Created: 30/05/2026

#3 ECO_TC_005 Hybrid: 56.8% Found in: both

Invalid Password Login Drop

Module: Auth High Risk: Medium Automated

Description: Validate that authentication drops if the password does not match system logs.

Steps: 1. Open sign-in panel 2. Enter valid email address 3. Enter incorrect password string 4. Click Submit

Expected Results: Access is denied with a visible credential mismatch error notification.

Score Breakdown:

BM25: 28.6% (raw: 3.00) Vector: 85.0% (raw: 0.85) Hybrid: 56.8%

Created: 30/05/2026

#4 ECO_TC_003 Hybrid: 53.2% Found in: both

Password Strength Boundary Validation

Module: Auth Medium Risk: Medium Automated

Description: Ensure the registration module enforces password complexity metrics.

Steps: 1. Go to register tab 2. Input an weak 4-character passcode 3. Attempt to submit form

Expected Results: Validation error triggers demanding at least 8 characters including symbols.

Score Breakdown:

BM25: 22.0% (raw: 2.31) Vector: 84.4% (raw: 0.84) Hybrid: 53.2%

Created: 30/05/2026

#5 ECO_TC_004 Hybrid: 53.1% Found in: both

Successful Buyer Login

Module: Auth High Risk: High Automated

Description: Confirm a registered buyer can authenticate cleanly using core credentials.

Steps: 1. Open login window 2. Enter valid username and password 3. Click Login

Expected Results: User is successfully authenticated and auth tokens are assigned to active session.

Score Breakdown:

BM25: 22.8% (raw: 2.39) Vector: 83.4% (raw: 0.83) Hybrid: 53.1%

Created: 30/05/2026

#6 ECO_TC_002 Hybrid: 51.8% Found in: both

Registration Duplicate Email Prevention

Module: Auth High Risk: High Automated

Description: Verify system blocks registration attempts using an already registered email profile.

Steps: 1. Navigate to sign up page 2. Enter an existing registered email 3. Fill in password fields 4. Click Register

Expected Results: System displays an error message stating email account is already in use.

Score Breakdown:

BM25: 20.9% (raw: 2.19) Vector: 82.8% (raw: 0.83) Hybrid: 51.8%

Created: 30/05/2026

#7 ECO_TC_001 Hybrid: 49.8% Found in: both

Successful User Sign Up

Module: Auth High Risk: High Automated

Description: Validate that a new user can successfully create an account with valid email and password.

Steps: 1. Navigate to the registration page 2. Enter a unique email address 3. Input a valid strong password 4. Click Register

Expected Results: The account should be created successfully and user redirected to the home screen profile.

Score Breakdown:

BM25: 16.0% (raw: 1.68) Vector: 83.5% (raw: 0.83) Hybrid: 49.8%

Created: 30/05/2026

#8 ECO_TC_050 Hybrid: 46.4% Found in: both

User Account Deletion Verification

Module: Auth High Risk: High Automated

Description: Confirm deleting an account purges user profiles completely out of active directories.

Steps: 1. Access secure user profile settings privacy option 2. Submit account deletion confirmation password instructions

Expected Results: Profile state changes to deleted user record is purged out of active directories and session closes.

Score Breakdown:

BM25: 8.8% (raw: 0.93) Vector: 84.1% (raw: 0.84) Hybrid: 46.4%

Created: 30/05/2026

#9 ECO_TC_040 Hybrid: 42.5% Found in: vector

MFA Verification Code Validation

Module: Auth High Risk: High Automated

Description: Ensure multi-factor authentication (MFA) layer challenges logins successfully.

Steps: 1. Attempt account authentication login from unknown terminal browser 2. Receive system generated MFA security text 3. Enter accurate verification token sequence

Expected Results: Security engine confirms token validation pattern passing access to profile window.

Score Breakdown:

BM25: 0.0% (raw: 0.00) Vector: 85.0% (raw: 0.85) Hybrid: 42.5%

Created: 30/05/2026

#10 ECO_TC_006 Hybrid: 42.3% Found in: vector

Account Lockout Threshold

Module: Auth Medium Risk: Medium Automated

Description: Verify account locks out temporarily after 5 sequential invalid sign-in attempts.

Steps: 1. Input bad passcode strings 5 consecutive times on login screen 2. Try a 6th login with true credentials

Expected Results: System temporarily locks the buyer profile and instructs them to try again in 15 minutes.

Score Breakdown:

BM25: 0.0% (raw: 0.00) Vector: 84.7% (raw: 0.85) Hybrid: 42.3%

Created: 30/05/2026

Reranking

1. In the frontend ui select 'Score Fusion' in the navigation tool
2. Under search query text box enter query 'reset password using otp' and click search

After ReRanking

Home > Score Fusion

BM25+Vector fusion

Groq AI Reranking Search

Combines BM25 keyword search and Vector semantic search, then uses Groq AI for intelligent semantic reranking and relevance optimization.

Top-K ...
50

Final ...
10

SEARCH

[FILTERS](#)

Groq AI Reranking Results: Found 10 test cases ranked by Groq AI
 Timing: Total: 6474ms
 Method: BM25 + Vector → Groq AI Semantic Reranking

AFTER RERANKING (FINAL)	BEFORE RERANKING (ORIGINAL)	COMPARISON TABLE
After Groq AI Reranking (10 results) #1 ECO_TC_007 Rerank Score: 1.0000 Was #1 Reset Password OTP Token Generation Module: Auth High Risk: High Description: Ensure password reset triggers an immediate temporary passcode token via email. Ranking Scores: BM25 Score: 10.4863 ★ Groq Rerank: 1.0000		

Before Ranking

[Home](#) > [Score Fusion](#)

BM25+Vector fusion results



Groq AI Reranking Search

Combines BM25 keyword search and Vector semantic search, then uses Groq AI for intelligent semantic reranking and relevance optimization.

Top-K ...

Final ...



Groq AI Reranking Results: Found 10 test cases ranked by Groq AI
Timing: Total: 6474ms
Method: BM25 + Vector → Groq AI Semantic Reranking

AFTER RERANKING (FINAL)

BEFORE RERANKING (ORIGINAL)

COMPARISON TABLE


Initial Candidate Ranking (Before Groq AI Reranking - 10 results from 50 candidates)



These are the top candidates from the 50-candidate pool before Groq AI semantic reranking. Notice how Groq reorders them based on actual relevance to your query!

Candidate Rank: #1

ECO_TC_007

Vector: 0.0000

Groq Score: 1.0000

Reset Password OTP Token Generation

Module: Auth

High

Description: Ensure password reset triggers an immediate temporary passcode token via email.

Candidate Rank: #2

ECO_TC_008

Vector: 0.0000

Groq Score: 0.9500

Comparison table

Groq AI Reranking Results: Found 10 test cases ranked by Groq AI

Timing: Total: 6474ms

Method: BM25 + Vector → Groq AI Semantic Reranking

AFTER RERANKING (FINAL)

BEFORE RERANKING (ORIGINAL)

COMPARISON TABLE

Before vs After Comparison

Original Rank (Candidates)	Test Case ID	Title	Vector Score	Groq Rerank Score	Final Rank
#1	ECO_TC_007	Reset Password OTP Token Generation	0.0000	1.0000	#1
#2	ECO_TC_008	Expired OTP Password Validation	0.0000	0.9500	#2
#9	ECO_TC_040	MFA Verification Code Validation	0.8497	0.7000	#3
#3	ECO_TC_005	Invalid Password Login Drop	0.0000	0.4000	#4
#5	ECO_TC_003	Password Strength Boundary Validation	0.0000	0.3500	#5
#6	ECO_TC_002	Registration Duplicate Email Prevention	0.0000	0.3000	#6
#7	ECO_TC_001	Successful User Sign Up	0.0000	0.2500	#7
#4	ECO_TC_004	Successful Buyer Login	0.0000	0.2000	#8
#8	ECO_TC_050	User Account Deletion Verification	0.0000	0.1500	#9
#10	ECO_TC_006	Account Lockout Threshold	0.8470	0.1000	#10

How to read this table:

• **Original Rank:** Position in the 50-candidate pool

• **Vector Score:** Semantic similarity (0.0-1.0)

• **Groq Rerank Score:** AI-based relevance score

• **Final Rank:** Top-10 position after reranking

• **Highlighted rows** = Top-3 results by Groq AI

• Compare Original Rank vs Final Rank to see how much Groq reordered each result

Search Summarization & Deduplication

1. In the frontend ui select 'Summarize and dedup' in the navigation tool
2. Under search query text box enter query 'reset password using otp' and click search

Home > Summarize & Dedup

AI summarization & deduplication

Search Summarization & Deduplication

Search, remove duplicates, and generate AI-powered summaries of test cases

Search Query

password reset OTP

Search Type

Vector

Limit

20

Quick Examples:

patient login authentication

merge UHID records

appointment booking

password reset OTP

Search

Deduplicate

Summarize with AI

Copy all results

Dedup Threshold

0.85

Summary Type

Concise

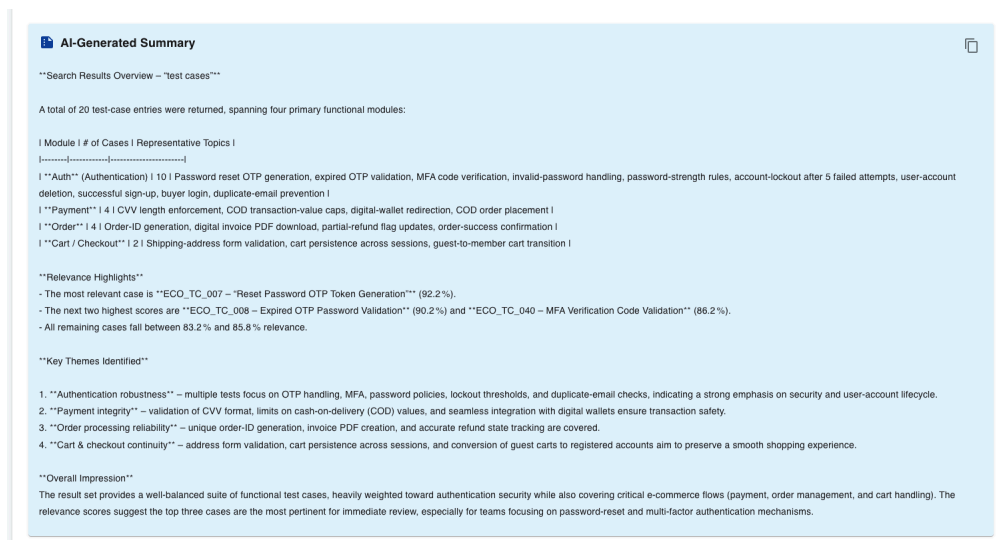
Show Duplicates

Original Results (20)	
ECO_TC_007 Score: 0.9222 Reset Password OTP Token Generation Ensure password reset triggers an immediate temporary passcode token via email. AuthHigh PriorityFunctional	ECO_TC_008 Score: 0.9021 Expired OTP Password Validation Verify that an expired password token fails validation during security resets. AuthMedium PriorityFunctional
ECO_TC_040 Score: 0.8618 MFA Verification Code Validation Ensure multi-factor authentication (MFA) layer challenges logins successfully. AuthHigh PriorityFunctional	ECO_TC_005 Score: 0.8577 Invalid Password Login Drop Validate that authentication drops if the password does not match system logs. AuthHigh PriorityFunctional
ECO_TC_003 Score: 0.8572 Password Strength Boundary Validation Ensure the registration module enforces password complexity metrics. AuthMedium PriorityFunctional	ECO_TC_006 Score: 0.8568 Account Lockout Threshold Verify account locks out temporarily after 5 sequential invalid sign-in attempts. AuthMedium PriorityFunctional
ECO_TC_050 Score: 0.8492 User Account Deletion Verification Confirm deletion of account removes user profiles completely, not of active directory. AuthMedium PriorityFunctional	ECO_TC_001 Score: 0.8410 Successful User Sign Up Validate that a new user can successfully create an account with valid email and password. AuthMedium PriorityFunctional

Click on DeDuplicate button

SEARCH RESULTS 20		DEDUPLICATED	AI SUMMARY								
Deduplication Statistics <table><tr><th>Original Count</th><th>After Deduplication</th><th>Duplicates Removed</th><th>Reduction</th></tr><tr><td>20</td><td>20</td><td>0</td><td>0.0%</td></tr></table>				Original Count	After Deduplication	Duplicates Removed	Reduction	20	20	0	0.0%
Original Count	After Deduplication	Duplicates Removed	Reduction								
20	20	0	0.0%								
Unique Results (20)											
ECO_TC_007 ✓ Unique Score: 0.9222 Reset Password OTP Token Generation Ensure password reset triggers an immediate temporary passcode token via email. AuthHigh PriorityFunctional	ECO_TC_008 ✓ Unique Score: 0.9021 Expired OTP Password Validation Verify that an expired password token fails validation during security resets. AuthMedium PriorityFunctional										
ECO_TC_040 ✓ Unique Score: 0.8618 MFA Verification Code Validation Ensure multi-factor authentication (MFA) layer challenges logins successfully. AuthHigh PriorityFunctional	ECO_TC_005 ✓ Unique Score: 0.8577 Invalid Password Login Drop Validate that authentication drops if the password does not match system logs. AuthHigh PriorityFunctional										

Click on Summarize with AI



1. How Reranking Works

Reranking takes your initial 50 search results and uses an AI model (Cross-Encoder) to re-score them based on relevance to your query. It's like a second opinion—the first search gives you candidates, the reranker intelligently reorganizes them to put the best matches at the top. Documents can jump up or down in ranking depending on how relevant the AI model thinks they are.

2. How Duplicate Results Are Identified and Removed

The system compares documents using a similarity score (0-1 scale). If two documents are 95% similar or more, they're considered duplicates. The system keeps the higher-ranked one and discards the duplicate. This prevents showing nearly identical test cases multiple times in results.

3. How Vector Search, BM25, and Hybrid Search Work

BM25 Search: Keyword-based matching. Searches for exact words in specific fields (like test ID, title, description) with heavier weight on more important fields. Good for finding specific test case IDs or exact terminology.

Vector Search: Semantic/meaning-based matching. Converts both your query and documents into numerical vectors (embeddings). Finds documents with similar meaning, even if exact keywords don't match. Good for conceptual searches like "patient communication flow" finding tests about notifications or messaging.

Hybrid Search: Combines both methods. Runs BM25 and Vector Search in parallel, then merges results using score fusion so you get the benefits of both keyword matching AND semantic understanding.

4. How Hybrid Weightages Impact Output

The weights control the balance between keyword matching (BM25) and semantic matching (Vector):

- 50/50 (Balanced): Both methods equally important—general-purpose search
- 70/30 (Keyword Heavy): Prioritizes exact matches—best for specific test case IDs like "TC_001"
- 30/70 (Semantic Heavy): Prioritizes meaning—best for broad concept searches like "authentication workflow"

Higher BM25 weight = exact matches bubble up. Higher Vector weight = conceptually similar results bubble up. This lets you tune results based on search intent.